

SECTION 01 35 20

SAFETY AND HEALTH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in the prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. Asbestos: 29 CFR 1926.1101.
- B. CIH: Certified Industrial Hygienist.
- C. Coatings containing Regulated Metals:
1. May be present in quantities that waste generated from removal operations may be a health hazard to employees. Other contaminants may be present.
 2. Metals and compounds that may be present in aging paint and coating systems:
 - a. Arsenic: 29 CFR 1926.1118.
 - b. Cadmium: 29 CFR 1926.1127.
 - c. Lead: 29 CFR 1926.6
- D. Competent Person:
1. 29 CFR 1926.32: Definitions.
 2. 29 CFR 1926.62: Lead.
 3. 29 CFR 1926.1101: Asbestos.
 4. 29 CFR 1926.1118: Arsenic.
 5. 29 CFR 1926.1127: Cadmium.
- E. MAP: EPA Asbestos Model Accreditation Program, defined in 40 CFR 763.
- F. NIOSH: National Institute of Occupational Safety and Health.
- G. NLLAP: National Lead Laboratory Accreditation Program.
- H. NVLAP: National Voluntary Laboratory Accreditation Program.
- I. PFT: Pulmonary Function Tests.
- J. PLM: Polarized Light Microscopy.

- K. PPE: Personal Protective Equipment.
- L. Regulated Area: See 29 CFR 1926.1101.
- M. TSP: Total Suspended Particulates.

1.03 REFERENCE STANDARDS

- A. Applicable Tribal and State Safety and Health Regulations for Construction.
- B. ASTM International (ASTM)
 - 1. ASTM D6480-19 Wipe Sampling of Surfaces, Indirect Preparation, and Analysis for Asbestos Structure Number Surface Loading by Transmission Electron Microscopy
 - 2. ASTM D6966-18 Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Determination of Metals
 - 3. ASTM D7035-16 Determination of Metals and Metalloids in Airborne Particulate Matter by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES)
 - 4. ASTM E1368-14 Visual Inspection of Asbestos Abatement Projects
- C. Bureau of Reclamation (Reclamation)
 - 1. RSHS Reclamation Safety and Health Standards including revisions posted at <https://www.usbr.gov/safety/rshs/index.html>
 - 2. FIST 1-1 (2012) Hazardous Energy Control Program available at: www.usbr.gov/power/data/fist_pub.html
- D. Code of Federal Regulations (CFR)
 - 1. 29 CFR 1926.32 Definitions, Construction
 - 2. 29 CFR 1926.55 Gases, vapors, fumes, dusts, and mists
 - 3. 29 CFR 1926.62 Lead
 - 4. 29 CFR 1926.103 Respiratory Protection
 - 5. 29 CFR 1926.1101 Asbestos
 - 6. 29 CFR 1926.1118 Inorganic Arsenic
 - 7. 29 CFR 1926.1127 Cadmium

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| 8. | 29 CFR 1926.1201-1213 | <i>Confined Spaces in Construction</i> |
| 9. | 40 CFR 763 | Asbestos |

E. Applicable State Safety and Health Regulations

1. California Code of Regulations (CCR) Title 8 – CAL/OSHA
 - a. Available online at: <https://www.dir.ca.gov/title8/index/T8index.asp>.
2. ***Deleted***

F. If conflict occurs between Federal and State Laws or standards, the most stringent having jurisdiction shall apply.

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 01 35 20-1, Safety Program:

1. Written Safety Program in accordance with Section 1.03 of the RSHS and WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).
2. Cover all aspects of onsite and applicable off-site operations and activities associated with this contract.
3. Program must follow the format of the Site-Specific Safety and Health Plan outline in Part 1.08.D below.
4. Will not be accepted for review by the COR unless it addresses, in order, lettered and numbered, in accordance with, the Site-Specific Safety and Health Plan outline in Part 1.08.D below.
 - a. For items in the outline that do not apply to this contract write “Not Applicable to Contract” next to the number/letter.
 - b. The outline is the minimum required safety elements. The Contractor may add elements to the outline that the Contractor believes are omitted but are relevant to the work being performed on the contract.
5. Provide a list of Job Hazard Analyses (JHA) for each distinct phase of work that will be submitted throughout the project.
6. A generic Company Safety Plan is not acceptable. The Safety Program must be site specific for this contract. Do not cut-and-paste parts from the RSHS.
7. The Contractor’s Safety Program must be submitted and accepted by Reclamation prior to commencing on site work, including mobilization.

C. RSN 01 35 20-2, Job Hazard Analysis List:

1. Provide a list of JHA’s that shall be submitted throughout the project.

- D. RSN 01 35 20-3, Asbestos Personnel Protection Plan:
1. In accordance with 29 CFR 1926.1101.
 2. Reviewed and signed by Competent Person.
 3. Include:
 - a. Compliance Program for Worker Protection:
 - 1) Respirator program as required by 29 CFR 1926.103 and 29 CFR 1926.1101.
 - 2) PFT for respirator usage: Record of diagnostic test and examination by a Board Eligible or Certified Occupational Medicine Physician.
 - 3) Manufacturers' data for safety and health equipment used in work. Include name, model, capacity, performance characteristics and other pertinent information.
 - a) Respirators.
 - b) Air Monitoring equipment.
 - c) PPE.
 - b. Employee Exposure Monitoring Plan:
 - 1) Monitoring airborne asbestos.
 - 2) Decontamination of contaminated areas and surfaces.
 - c. Clearance testing.
- E. RSN 01 35 20-4, Regulated Metals in Coatings Personnel Protection Plan:
1. Worker Protection Compliance Program:
 - a. Include information required by 29 CFR 1926.62.
 - b. Prepared by the Competent Person.
 - c. Exposure Certifications and Test Results:
 - 1) Employee training certifications: Indicate employee training in hazardous materials and communication with an emphasis on Regulated Metals, respiratory protection, and PPE.
 - 2) PFT for respirator usage: Record of diagnostic test and examination by a Board Eligible or Certified Occupational Medicine Physician.
 - 3) Blood tests for Regulated Metals:
 - a) Blood lead and zinc protoporphyrin level (ZPP).
 - b) Blood levels for other Regulated Metals.
 2. Employee Exposure Monitoring Plan:

- a. Plan for monitoring regulated metal levels from employee exposure personal monitors.
 - b. Signed by CIH.
- F. RSN 01 35 20-5, Asbestos Preconstruction Test Results:
 - 3. Background air monitoring results:
 - a. From personal employee exposure monitors.
 - b. Map indicating location of each sample.
 - c. Quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples with percent recovery, and other appropriate quality control measures.
- G. RSN 01 35 20-6, Asbestos Working Level Test Results:
 - 1. Personal Employee exposure and area monitoring results by personal employee exposure monitors.
 - 2. Compare with preconstruction and during construction test results for increased contaminant levels.
 - 3. Pressure differential recordings for containment area if used.
 - 4. Map indicating location of each sample.
 - 5. Quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples with percent recovery, and other appropriate quality control measures.
- H. RSN 01 35 20-7, Asbestos Post-Construction Test Results:
 - 1. Personal Employee exposure and area monitoring results by personal employee exposure monitors.
 - 2. Compare with preconstruction and during construction test results for increased contaminant levels.
 - 3. Map indicating location of each sample.
 - 4. Quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples with percent recovery, and other appropriate quality control measures.
- I. RSN 01 35 20-8, Regulated Metals Preconstruction Test Results:
 - 1. Background air monitoring results:
 - a. Personal Employee exposure and area monitoring results by personal employee exposure monitors.
 - b. Map indicating location of each sample.

- c. Quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples with percent recovery, and other appropriate quality control measures.
- J. RSN 01 35 20-9, Regulated Metals Working Level Test Results:
 - 1. Include:
 - a. TSP air quality monitoring results for Regulated Metals concentrations by personal employee exposure monitors.
 - b. Air quality monitoring results for TSP lead by PM-10.
 - 2. Map indicating location of each sample.
 - 3. Provide appropriate quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples from project with percent recovery, and other appropriate quality control measures.
- K. RSN 01 35 20-10, Regulated Metals Post-Construction Test Results:
 - 1. Personal Employee exposure and area monitoring results by personal employee exposure monitors.
 - 2. Blood tests for Regulated Metals:
 - a. Blood lead and zinc protoporphyrin level (ZPP).
 - b. Blood levels for other Regulated Metals.
 - 3. Compare with preconstruction test results for increased contaminant levels.
 - 4. Include map indicating location of each sample.
 - 5. Quality control information to supplement analytical data. From blanks, laboratory control samples with percent recovery, duplicates, matrix spike samples with percent recovery, and other appropriate quality control measures.
- L. RSN 01 35 20-11, Monthly Accident Summary Report:
 - 1. In accordance of RSHS.
- M. RSN 01 35 20-12, Documentation and Records.
 - 1. Submit documentation of medical qualifications less than 12 months old for all respirator users at the site for work not covered above.
 - 2. Submit results of fit tests less than 12 months old for all users of tight fitting negative pressure respirators at the site for work not covered above.

1.05 DOCUMENTATION AND RECORDS

- A. Prepare and retain all safety plans, programs, training content, and training records that are applicable to the scope of the work and make them available to the COR unless they

are already included in the written safety program. Some examples include but are not limited to:

1. Asbestos Exposure Assessment and Training Records.
2. Benzene Written Program and Training Records.
3. Cadmium Written Compliance Program and Training Records.
4. Chromium VI Training Records.
5. Confined Space and when deemed necessary by existing and/or introduced hazards of Permit Required Confined Space through the Written Program and Training Records.
6. Forklifts and Other Industrial Trucks Training Records.
7. Electrical Safety Requirement Training Records.
8. Emergency Action Plans Written Program and Training Records.
9. Fall Protection Written Program and Training Records.
10. Fire Protection and Prevention Written Program and Training Records.
11. Flammable and Combustible Liquids Written Spill Control Plan.
12. General Safety and Health Provisions, Written Safety Program, and Regular Safety Inspection Records.
13. Hazard Communication Written Program and Training Records.
14. Inorganic Arsenic Training Records, if applicable.
15. Ladder Written Compliance Program and Training Records.
16. Lead Written Program and Training Records.
17. Lockout/Tagout: Refer to RSHS Section 15 Hazardous Energy Control Program (HECP) Training Records.
18. Bloodborne Pathogens Exposure Control Plan, Training Records.
19. Methylenedianiline Training Records, if applicable.
20. Occupational Noise Exposure Written Program and Training Audiometric Testing Records.
21. Personal Protective Equipment (PPE) Hazard Assessments and Training Records.
22. Power Operated Hand Tools Training Records.
23. Respiratory Protection Written Program and Training Records.
24. Safety Training and Education Written Program and Training Records.
25. Scaffolds Training Records.
26. Steel Erection Training Records, if applicable.
27. Welding, Cutting, and Brazing Written Program and Training Records.

1.06 SAFETY AND HEALTH

- A. The Contractor shall not require any person employed in the performance of this contract (including subcontracts) to work under conditions which are unsanitary, hazardous, or dangerous to an employee's health or safety in accordance with the current version of WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).
 - 1. Do not allow Contractor employees (including subcontractors), Government personnel, or the Public to be exposed to, or work under, conditions which are unsanitary, physically hazardous, or a potential exposure danger to their health or safety during performance of the contract.
- B. The Contractor shall comply with the most current version of the Bureau of Reclamation's Reclamation Safety and Health Standards (RSHS) and, if incorporated, 52.236-13 Accident Prevention in accordance with WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).
- C. The Contractor shall comply with the standards (29 CFR) issued by the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, www.osha.gov in accordance with WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).
 - 1. In the event there is a conflict between the requirements contained in paragraphs (B) and (C) referenced herein, or in the Specifications, the more stringent requirement having jurisdiction shall prevail.
- D. Under no circumstances will onsite work, including mobilization, be permitted until the Safety Program has been accepted by the COR.
- E. Fully participate in a Contractor Safety Program Review meeting, according to RSHS Section 1.03, prior to mobilization. If applicable, include subcontractor management representatives.
- F. The minimum work crew at any time on the construction site shall consist of no less than two people and shall be in accordance with other contractual obligations.
- G. Develop Job Hazard Analysis for each distinct phase of work under the contract.
 - 1. Work shall not begin on the phase of work until a JHA is acceptable to COR and shared with all construction employees.
 - 2. Activities involving hazardous materials shall have the appropriate Safety Data Sheet(s) attached to the JHA.
- H. In addition to requirements of the clause at FAR 52.236-13 "Accident Prevention", fully comply with RSHS.
 - 1. RSHS is available online at <https://www.usbr.gov/safety/rshs/index.html>.
 - 2. Maintain onsite a copy of the accepted RSHS Safety Program.

3. Construction Safety and Health Standards promulgated by the Secretary of Labor may be obtained from any regional or area office of the Occupational Safety and Health Administration of the U.S. Department of Labor.
 4. California Code of Regulations (CCR) Title 8 Occupational Safety and Health Standards that apply to all (Non-Federal) places of employment in California.
 5. Maintain onsite a required for every California employer a copy of the written Injury and Illness Prevention (IIP) Program.
- I. Be cognizant of and ensure compliance with requirements set forth in paragraphs above.
1. Contractor's responsibility applies to all operations, including those of the Contractor's Subcontractors.
 2. When violations of safety and health requirements contained in these specifications or referenced standards are called to the Contractor's attention by the CO and COR immediately correct the condition to which attention has been directed.
 3. Either oral or written notice shall be deemed sufficient.
- J. When the Contractor fails or refuses to promptly correct a compliance directive, the CO or the COR may issue an order to stop all or any part of the work.
1. When satisfactory corrective action is taken, an order to resume work will be issued.
 2. The Contractor shall not be entitled to extension of time, nor to claim for damage or to additional compensation by reason of either the directive or the stop order.
 3. Failure of the CO and COR to order discontinuance of any or all of the Contractor's operations shall not relieve the Contractor of the responsibility for the safety of personnel and property.
- K. The Contractor shall maintain an accurate record of, and shall report to the Contracting Officer (or authorized representative) in the manner prescribed by the Contracting Officer, all cases of death, occupational diseases, or traumatic injury to employees or the public involved, and property damage in accordance with the RSHS, OSHA guidelines, and with WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).
1. The Contractor shall immediately notify the Construction Office Safety Manager following the reporting to OSHA of a work-related OSHA-Reportable case including, fatality, inpatient hospitalization, amputation, or loss of an eye.
 2. Maintain an accurate record of, and report to the COR in the manner prescribed by the CO for all near misses, accidents or incidents, occupational diseases, traumatic injury, cases of death to Contractor employees (including subcontractors), Government personnel, or the Public, and any property damage occurring during the performance of work under this contract.

- L. The rights and remedies of the Government provided in this section are in addition to any other rights and remedies provided by law or under this contract.
1. In the event there is a conflict between requirements contained in RSHS, specification paragraphs, Contractor's approved Safety Program, referenced safety and health codes, and standards, or the U.S. Department of Labor Construction Safety and Health Standards, promulgated under Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 327 et seq.), as amended, the more protective requirement having jurisdiction shall prevail.
- M. If the contractor deems that an updated version of the RSHS constitutes a substantive change to the contract, affecting price or time (or both), the contractor may request a contract modification in accordance with the terms of the contract in accordance with WBR 1452.223-81-- SAFETY AND HEALTH -- BUREAU OF RECLAMATION (OCT 2022).

1.07 Deleted

1.08 Deleted

1.09 SAFETY TRAINING VISUAL DOCUMENTATION

- A. Provide visual documentation of completed safety trainings (i.e. hard hat stickers).
1. The visual documentation will enable people to be identified who have or have not received quarry safety training.
2. Quarry safety training is required for access into the quarry area unless escorted by an individual who has been trained.

1.10 SAFETY AND HEALTH PLAN

- A. Unless waived by the Contracting Officer's Representative (COR), prime and supplementary safety program submittals shall, as a minimum, address all appropriate contractual requirements shown under the program outline. Program items will be so detailed that reviewers can ascertain adequacy. The original program submittal can cover all work phases, or only address the initial work phase with timely supplementary submittals as major work phases occur. Contractors desiring to follow this latter course of action will, in the initial submission, clearly define the original work covered and the work phases to be covered by supplements.
- B. In no case will any phase of work commence until a program for that portion of the work has been accepted by the COR. Original and supplemental submissions covering hazardous operations and/or activities will include a standing operating procedure (SOP) and hazard analysis. The SOP will break down the operation or activity into specific basic steps. The hazard analysis will define the hazards associated with each basic step and proposed method(s) for eliminating or minimizing the hazard. Such methods will outline, as a minimum, employee training requirements, personal protective equipment requirements, procedural changes, methods for evaluating program effectiveness, etc.

- C. All activities involving use of hazardous and/or toxic materials, work in confined spaces, work at heights over 6 feet, or underwater require an SOP and hazard analysis. Material Safety Data Sheets will be attached to all SOPs governing work that involves use of hazardous or toxic materials.
- D. Prepare a Site-Specific Safety and Health Plan with major components labeled in accordance with the following Program Outline:
 - 1. General Requirements:
 - a. Statement of Policy.
 - b. Statement of Safety and Health Responsibilities.
 - c. Statement of Compliance with Regulations, Standards, and Codes.
 - d. Statement of Subcontractor Compliance.
 - e. Safety Inspection Procedures.
 - f. Accident Investigation and Reporting Procedures.
 - g. Applicable Emergency Plans.
 - h. Confined Space Procedures.
 - i. Lockout/Tagout Procedures.
 - j. Fire Protection Plans.
 - k. Type and location of suppression equipment or systems.
 - l. Offsite assistance agreement.
 - m. Temporary heating devices.
 - 2. Medical:
 - a. Facilities.
 - b. Training.
 - c. Certifications.
 - d. Physician.
 - e. Ambulance (Name, location, and telephone number).
 - f. Physical Qualification of Employees.
 - g. Records.
 - 3. Communications:
 - a. Employee Training.
 - b. Safety Meetings.
 - c. Onsite Training.
 - d. Supervisor Training.

4. Occupational Health:
 - a. Procedures and Equipment to Minimize Hazards.
 - b. Testing program for employees and work environments.
 - c. Qualified personnel.
 - d. Personal protective equipment.
 - e. Ventilation plans.
5. Machinery and Mechanical Equipment:
 - a. Procedures and Equipment to Minimize Hazards.
 - 1) Testing program for employees and work environments.
 - 2) Mobile and stationary equipment.
 - b. Inspection Procedures.
 - c. Maintenance Procedures.
 - d. Operating Personnel.
 - e. Protective Safety Devices and Certifications.
 - f. Elevators and Aerial Lifts.
6. Excavation and Demolition:
 - a. Tunnels and Shafts:
 - 1) Internal combustion engines.
 - 2) Ventilation plans.
 - 3) Transportation systems and equipment.
 - 4) Work environment testing.
 - 5) Ground support.
 - b. Blasting:
 - 1) Blaster certification.
 - 2) Written procedures.
 - 3) Storage.
 - 4) Transportation.
 - c. Excavations Other Than Tunnels and Shafts:
 - 1) Slide protections.
 - 2) Support systems.
 - 3) Inspections.
 - 4) Access.
 - d. High Scaling:

- 1) Definition.
 - 2) Personal protective equipment.
 - 3) Standing operating procedure.
 - e. Haulage:
 - 1) Haul roads.
 - 2) Equipment and Procedures.
7. Working Surfaces:
 - a. Access:
 - 1) Ladders.
 - 2) Platforms, stairways, and ramps.
 - b. Personal Protective Equipment.
 - c. Scaffolding.
 - d. Safety Nets.
8. Protection of the Public:
 - a. Signs and Barricades.
 - b. Flagging Procedures.
 - c. Jurisdictional Approvals.
9. Marine and Diving Operations:
 - a. Detailed Plan and Written Procedures.
10. Electrical Facilities:
 - a. Substations.
 - b. Required Safety Program Coordination:
 - 1) Confined Space Program.
 - 2) Electrical and Lockout/Tagout Program.
11. Fall Protection:
 - a. Worksite Hazard Analysis.
 - b. System Selection.
 - c. System Installation.
 - d. System Inspection.
 - e. Rescue Planning.
 - f. Fall Protection Training.

- E. This outline provides guidance in preparing a safety program and does not cover all material that may be necessary. The Contractor shall review specifications and all safety and health regulations to ensure a comprehensive plan.

1.11 CONFINED SPACES

- A. *All areas in the outlet works tunnel and powerplant turbine room are confined spaces.*
- B. *The following locations have a potential engulfment hazard and are classified as Permit Required Confined Spaces:*
1. *Inside the 24-inch, 42-inch, and 90-inch outlet works pipes.*
 2. *All areas inside the outlet work tunnel and headgate chamber when the 90-inch outlet work access cover or any component of the 12-inch or 14-inch air-vac pipes are not fully installed.*
 3. *All areas inside the turbine room when the 24-inch butterfly valve, 42-inch butterfly valve, or 20-inch manhole access hatch to the 42-inch penstock are not fully installed.*
- C. *For permit required confined spaces, comply with requirements of 29 CFR 1926 Subpart AA; CalOSHA Title 8, Section 5157; and RSHS Section 1.12.*

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 AIR QUALITY MONITORING BY EMPLOYEE EXPOSURE PERSONNEL MONITORS

- A. Perform employee exposure monitoring to determine exposure to Regulated Metals in accordance with NIOSH or OSHA analytical sampling methods.
- B. Not to exceed emissions limit standards in:
1. 29 CFR 1926.55.
 2. 29 CFR 1926.62.
 3. 29 CFR 1926.1118.
 4. 29 CFR 1926.1127.
- C. At the same time that employee exposure monitoring is done, monitor airborne contaminants at Regulated Area boundary.

- D. If emissions exceed specified maximum allowable concentrations, stop work until containment system meets required standards.

END OF SECTION

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